Product Specification Document

Service Mobility Dashboard

Full-Stack Final Project

Product Overview

Service Mobility Dashboard is a web-based decision-support system designed for organizations that operate field-service vehicles and manage their service tasks through SAP.

The product provides service dispatchers with a unified geographical view of scheduled service tasks, optimized driving routes for each service vehicle, and decision-support tools for assigning new service tasks.

The system is not intended to replace SAP as the organization's source of truth. Instead, it serves as a lightweight operational layer that transforms existing task data into a clear, visual, and actionable planning tool.

What Problem Does the Product Solve?

Currently, planning the daily work of field-service vehicles is often performed manually and across several disconnected tools.

Service dispatchers must move between operational data stored in SAP and external tools such as calendars and mapping applications in order to understand where tasks are located, which vehicle is assigned to each task, and how the daily schedule should be organized.

Although SAP contains the required operational information, the raw data alone does not provide dispatchers with a clear geographical overview of the day's activity.

This working method creates several problems:

- **Time-consuming planning:** Dispatchers spend valuable time manually combining information from different systems in order to build a daily operational picture.
- **Lack of geographical visibility:** It is difficult to quickly understand how service tasks are geographically distributed between vehicles.
- **Inefficient driving routes:** Vehicles may travel unnecessary distances, revisit the same areas, or perform tasks in an inefficient geographical order.
- **Difficulty assigning new service calls:** When a new task arrives, the dispatcher must manually determine which vehicle and which upcoming workday would be the most geographically suitable assignment.
- **Slow decision-making:** Without a unified operational view, making quick scheduling decisions requires repeatedly checking several systems.
- **Information gaps and human errors:** Working in parallel with SAP, calendars, and maps increases the chance of missed information, duplicated work, or inconsistent planning.

The Service Mobility Dashboard addresses these problems by transforming the organization's service-task data into a centralized visual planning environment.

Instead of manually reconstructing the operational picture, the dispatcher can immediately see the relevant service tasks, their locations, their assigned vehicles, and optimized routes on one interactive map.

In addition, when a new service request is received, the dispatcher can search for its address and receive recommendations for vehicles that already have scheduled tasks near that location during the upcoming days. This allows the dispatcher to make a faster and more informed assignment decision.

Who Are the Users of the Product?

Primary Users - Service Dispatchers

The primary users are service dispatchers responsible for planning and coordinating the daily activity of field-service vehicles.

Their responsibilities include:

- Reviewing scheduled service calls.
- Understanding the geographical distribution of daily tasks.
- Coordinating work between different service vehicles.
- Evaluating where new service calls should be assigned.
- Monitoring customer addresses, time windows, and operational information.
- Making fast scheduling decisions throughout the workday.

The dashboard is designed primarily around the dispatcher's workflow and aims to reduce the number of systems and manual steps required to perform these tasks.

Secondary Users - Operations Managers

Operations managers may also use the system to obtain a high-level overview of field activity and understand how vehicles and service calls are distributed.

Indirect Stakeholders

Field technicians and end customers benefit from the decisions made through the system, although they are not necessarily direct users of the dashboard.

Technicians benefit from more efficient daily routes, while customers may benefit from improved scheduling and better adherence to service windows.

Who Is the Client?

The client is a **medium-to-large organization that operates field-service teams or vehicle fleets and already manages its core service operations through SAP**.

Examples include:

- Telecommunications companies.
- Maintenance and repair organizations.
- Infrastructure companies.
- Field technician organizations.
- Companies providing on-site installation services.
- Organizations performing complex field deliveries or service visits.

These organizations already possess the operational data required for planning, but need a lightweight visual and analytical layer that makes the information easier for dispatchers to use in day-to-day decision-making.

The organization itself is the client of the product. The end customers receiving the field service are not direct users of the system.

What Are the Business Goals of the Product?

1. Reduce Daily Planning Time

Significantly reduce the amount of time dispatchers spend collecting information from different systems and manually planning the daily activity of service vehicles.

The dispatcher should be able to obtain the relevant operational picture from a single interface.

2. Improve Assignment Decisions for New Service Tasks

Help dispatchers decide where a newly received service task should be scheduled.

Instead of manually examining several vehicles and future schedules, the system identifies existing service tasks located near the new address and presents possible vehicle-and-date combinations.

This provides the dispatcher with useful geographical information before making the final scheduling decision.

3. Reduce Unnecessary Travel

Improve the geographical order of service stops and reduce unnecessary driving between customer locations.

More efficient routes can contribute to:

- Reduced fuel consumption.
- Reduced vehicle wear.
- Reduced travel time.
- Fewer unnecessary trips between geographical areas.

4. Increase Operational Productivity

Reducing travel and planning overhead allows service teams to spend a larger portion of the workday performing actual service tasks.

Over time, this can help the organization make better use of its existing fleet and workforce.

5. Create a Unified Operational Picture

Provide dispatchers with one clear geographical view that combines information that would otherwise need to be reconstructed manually from SAP and external tools.

This reduces dependence on parallel planning tools and lowers the risk of information gaps and human errors.

6. Improve Service Quality

Better planning and reduced travel time can help the organization adhere more effectively to customer service windows and respond more efficiently to new service requests.

7. Preserve SAP as the Organizational Source of Truth

The dashboard is designed as a decision-support and visualization layer rather than a replacement for SAP.

Service-task data is treated as read-only inside the dashboard. This prevents the creation of conflicting versions of operational information and ensures that SAP remains the authoritative system for service-task management.

What Software Capabilities Are Required to Support the Business Goals?

To achieve the business goals, the product requires the following capabilities:

Secure Authentication

Only authorized organizational users should be able to access operational service information.

The application therefore requires secure login and logout functionality.

Daily Task Retrieval

The system must retrieve the service tasks scheduled for a selected date and present only the relevant information to the dispatcher.

Interactive Geographic Visualization

Service tasks must be displayed on an interactive map according to their geographical coordinates.

Tasks belonging to different vehicles should be visually distinguishable.

Vehicle-Based Task Grouping

Tasks must be grouped according to their assigned service vehicle so that dispatchers can quickly understand the daily workload and geographical activity of each vehicle.

Route Optimization

For vehicles with multiple service locations, the system should calculate and display an efficient driving order between the stops.

Date Navigation

Dispatchers must be able to move between different workdays and review scheduled service activity for past or future dates.

Detailed Task Information

The dispatcher must be able to select a service stop and immediately view its relevant operational details, such as:

- Customer address.
- Customer name.
- Customer phone number.
- Assigned vehicle.
- Assigned installer.
- Service time window.
- Operational notes.

This reduces the need to return to SAP simply to retrieve basic information while planning.

New Address Search

The system must allow the dispatcher to search for the address of a potential new service task and display its geographical location.

Vehicle and Date Recommendations

After selecting the address of a new task, the system should examine scheduled tasks during the upcoming four days.

It should identify vehicle-and-date combinations that already contain a service stop within a **20 km radius** of the new address.

For each vehicle and date, the closest existing stop is used as the relevant candidate.

The recommendations are then ordered by geographical distance, allowing the dispatcher to quickly identify potentially efficient scheduling options.

The recommendation is a **decision-support tool**. The dispatcher remains responsible for making the final scheduling decision.

Bilingual User Interface

The product should support both **English and Hebrew**, including right-to-left layout when Hebrew is selected, so that the dashboard can be comfortably used in the organization's operational environment.

Graceful Error Handling

Failures in an external routing service should not prevent the dispatcher from accessing the underlying service-task information.

If route optimization is unavailable, the system should still display the service stops on the map and allow the user to continue working.

What Are the Main Processes the Product Allows Users to Perform?

1. Secure Login

An authorized employee enters the application and signs in using organizational credentials.

After successful authentication, the employee receives access to the operational dashboard.

There is no public self-registration flow because the system contains internal organizational information and is intended only for authorized employees.

---

2. Select a Workday

The dispatcher can select a specific date using the dashboard's date navigation.

The system loads the service tasks scheduled for the selected day.

This allows the dispatcher to review today's activity as well as examine other planned workdays.

3. View the Daily Operational Map

The dispatcher sees all service tasks scheduled for the selected date on a centralized interactive map.

Tasks are grouped by their assigned vehicle, allowing the dispatcher to quickly understand:

- Which vehicles are active.
- Which stops belong to each vehicle.
- Where the daily activity is geographically concentrated.
- How different vehicle routes relate to one another.

4. Review an Optimized Route for Each Vehicle

For each vehicle with multiple valid service locations, the system calculates an optimized driving route between its stops.

The resulting route is displayed directly on the map together with numbered service-stop markers.

This gives the dispatcher an immediate visual understanding of the suggested order of travel.

5. Inspect a Specific Service Task

The dispatcher can select a stop either from the map or from the vehicle's task list.

The map focuses on the selected location and displays the relevant service information without requiring the dispatcher to leave the dashboard.

6. Search for a New Service Address

When a new service request is received, the dispatcher can enter its address into the dashboard.

The system converts the address into a geographical location and displays it on the map.

7. Receive Scheduling Recommendations

After a new address has been selected, the system examines service tasks already scheduled during the next four days.

It searches for vehicles that already have a scheduled stop within 20 km of the new location.

The dispatcher receives a list of candidate assignments containing information such as:

- Proposed workday.
- Distance from the new task to a nearby existing stop.
- Assigned installer.
- Existing service time window.

The recommendations are sorted by distance so that the most geographically relevant options appear first.

8. Compare a Recommendation With the Existing Schedule

The dispatcher can select one of the recommendations and move directly to that workday's dashboard view.

This allows the dispatcher to inspect the complete route and surrounding tasks before deciding whether the suggested assignment is operationally appropriate.

The system assists the decision but does not automatically modify the SAP schedule.

---

9. Switch the Interface Language

The dispatcher can switch between English and Hebrew.

When Hebrew is selected, the application adapts the interface to a right-to-left layout.

---

10. Logout

When the dispatcher finishes working with the system, they can securely log out and end the authenticated session.

Product Scope and Boundaries

The current product is designed primarily as a **visualization and decision-support layer** for service dispatchers.

The system:

- Reads service-task information that originates from SAP.
- Displays and organizes that information geographically.
- Calculates optimized routes.
- Helps dispatchers evaluate potential assignments for new service requests.

The system does **not** currently:

- Replace SAP as the organization's main operational system.
- Directly modify service-task records in SAP.
- Automatically assign a new task to a vehicle.
- Automatically change the operational schedule based on a recommendation.

These boundaries are intentional. The goal of the current version is to provide dispatchers with better information and faster decision-support while keeping SAP as the single authoritative source for operational data.